

Disentangling and Integrating Relational and Sensory Information in Transformer Architectures

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Motivation & High-level Goals

Relational reasoning is fundamental to intelligence

- Cornerstone of human intelligence, underlying capabilities for *analogy*, *abstraction*, *generalization*
- Standard Transformers are data-inefficient at relational tasks and have brittle OOD generalization
- Prior relational architectures are narrow in domain due to stringent subtractive inductive biases.

Our Goal: Augment Transformers with explicit relational mechanisms while retaining sensory processing capabilities

Key Insight: Two types of information require two types of attention:

- **Sensory** : features of individual objects
- **Relational** : relationships between objects

Transformers: The Sensory & The Relational

Strength of Transformers: *attention*; Versatile *information retrieval* mechanism

The Transformer architecture, essentially:

1. **Information Retrieval:** Attention

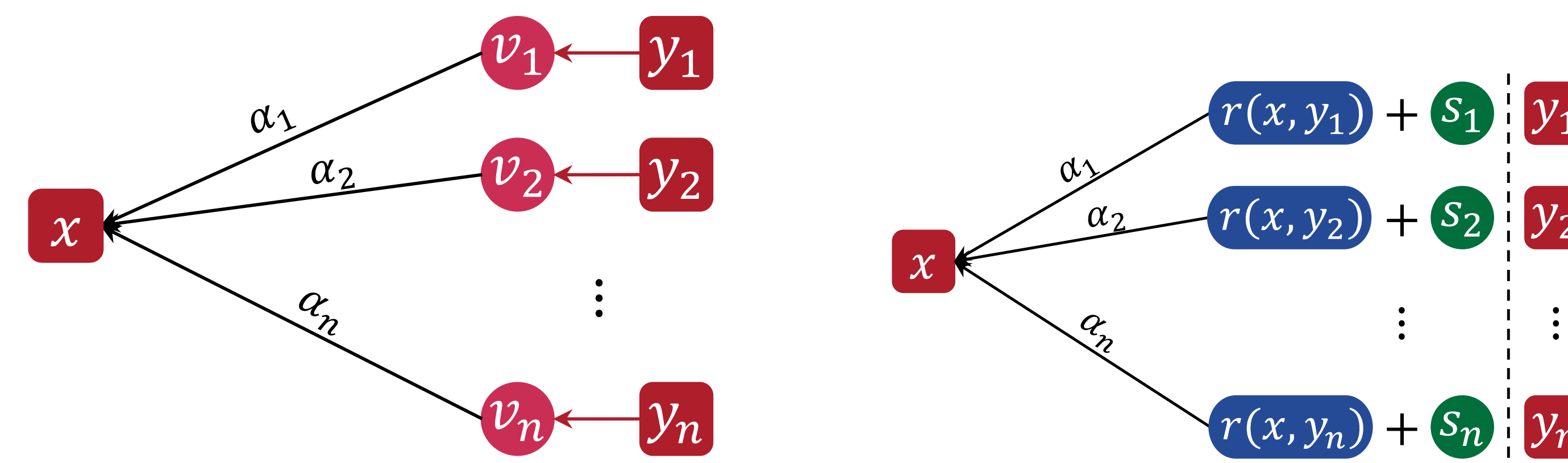
$$x'_i \leftarrow \sum_j \alpha_{ij} \phi_v(x_j)$$

2. **Local Processing:** Token-wise MLP

$$x'_i \leftarrow \text{MLP}(x_i)$$

3. Repeat

Disentangling Attention over Sensory & Relational Information



Left: **Sensory attention** retrieves object features. Right: **Relational attention** retrieves relation vectors tagged with **symbols**

Relational Attention: Three-step process

1. Attend: Compute attention scores

$$\alpha_{ij} = \text{Softmax}([\langle \phi_q^{\text{attn}}(x_i), \phi_k^{\text{attn}}(x_j) \rangle]_{j=1}^n)_j$$

2. Relate: Compute **relation vectors**

$$r_{ij} = (\langle \phi_{q,\ell}^{\text{rel}}(x_i), \phi_{k,\ell}^{\text{rel}}(x_j) \rangle)_{\ell \in [d_r]} \in \mathbb{R}^{d_r}$$

3. Tag with Symbols: Abstract **identifiers**

$$(s_1, \dots, s_n) = \text{SymbolRetriever}(x_1, \dots, x_n)$$

Relational Attention Output:

$$a_i \leftarrow \sum_j \alpha_{ij} \cdot (W_r r_{ij} + W_s s_j)$$

Integrating Attention over Sensory & Relational Information

Dual Attention Transformer (DAT): Multi-head attention with two types of heads

- **Sensory heads** : Standard self-attention
- **Relational heads** : Relational attention

Architecture variants:

- Encoder-only (vision tasks)
- Decoder-only (language modeling)
- Encoder-decoder (seq2seq tasks)

Experiments & Results

Synthetic **Relational** Tasks (Relational Games)

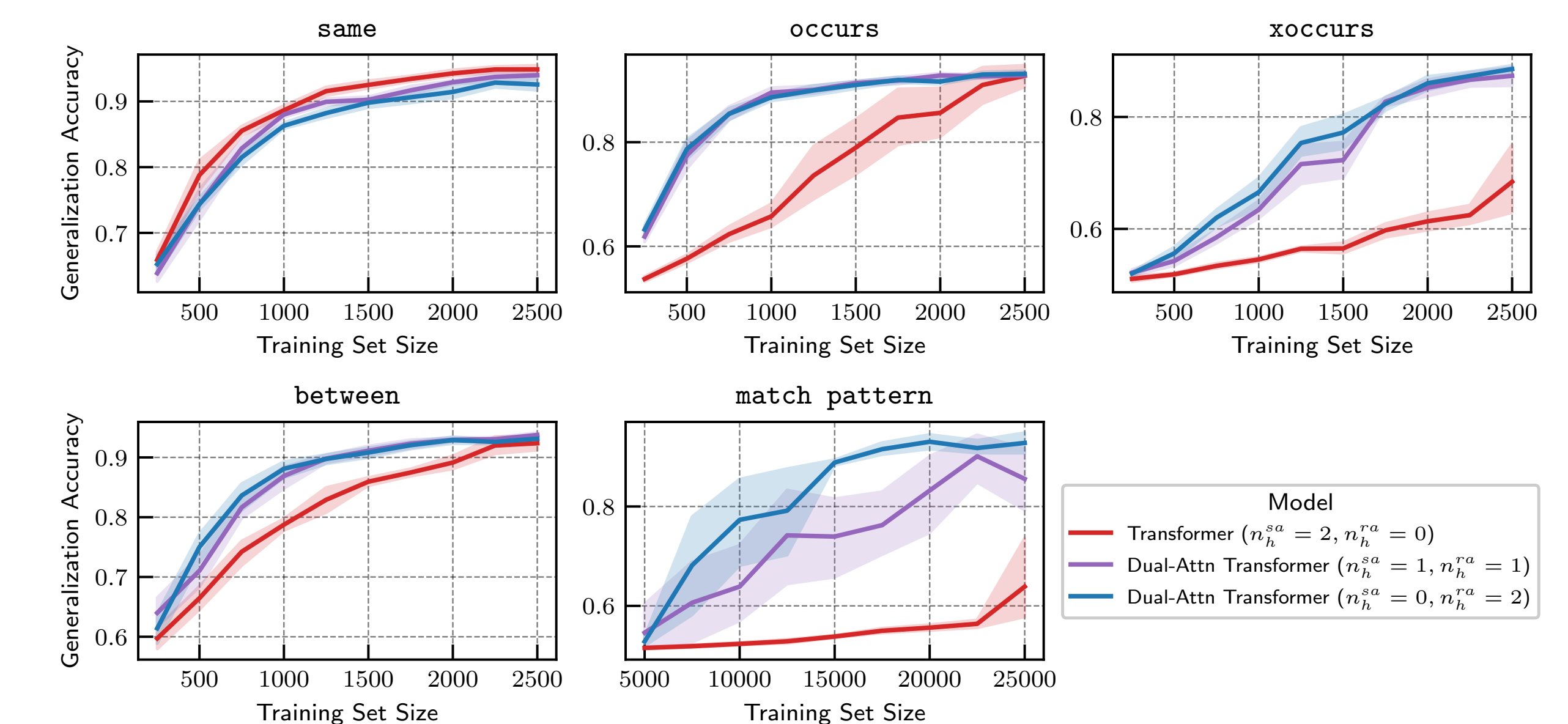


Figure 1. DAT significantly outperforms standard Transformers on synthetic relational reasoning tasks

Mathematical Problem Solving (Seq2Seq)

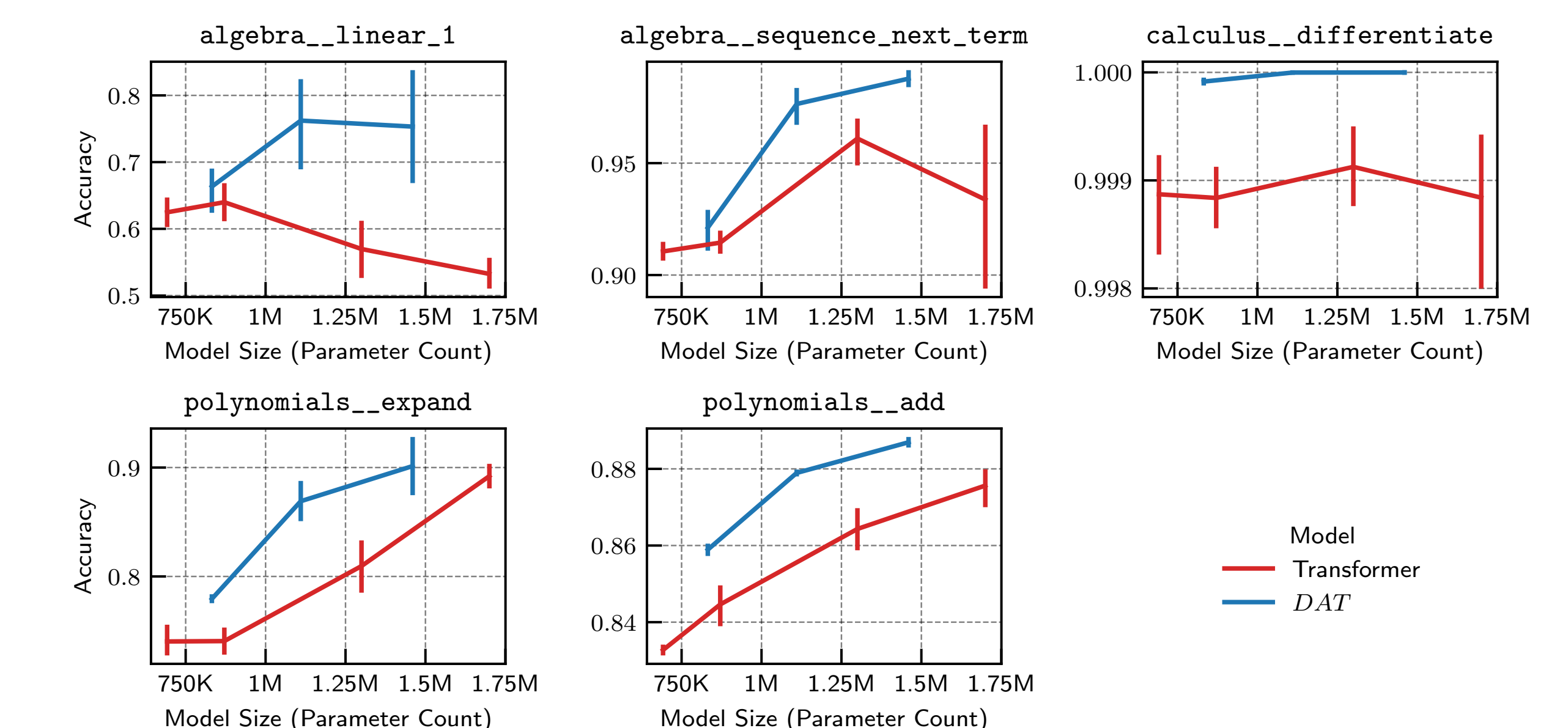


Figure 2. DAT shows superior scaling with model size on mathematical reasoning tasks

Vision Tasks (CIFAR)

Dataset	Model	Params	Accuracy
CIFAR-10	ViT	7.1M	86.4 ± 0.1%
	ViDAT	6.0M	89.7 ± 0.1%
CIFAR-100	ViT	7.2M	68.8 ± 0.2%
	ViDAT	6.1M	70.5 ± 0.1%

Table 1. Classification accuracy on CIFAR-10/100. DAT outperforms Vision Transformers

Language Modeling